EXPERIMENT NO. 3

Student Management using JSP, JDBC and MySQL

Final 8-file version discussed for your official Java and Spring Framework Lab question.

Required operations
Create - insert student details
Read - display all students
Update - modify an existing student
Delete - remove a student by roll number

Student fields: roll number, name, email and age.

Output style: simple labels, line breaks and horizontal separators. No HTML tables are used.

Important: Java classes were compile-checked. The full JSP application has not been run against a live Eclipse/Tomcat/MySQL environment; perform a short end-to-end test before the practical.

1. Project structure in Eclipse

Use this structure because your project has `src/main/webapp`, not `WebContent`.

```
StudentApp/
  src/main/java/
    com/student/model/Student.java
    com/student/util/DBUtil.java
    com/student/dao/StudentDAO.java
  src/main/webapp/
    index.jsp
    add-student.jsp
    list-student.jsp
    update-student.jsp
    delete-student.jsp
    WEB-INF/lib/ (MySQL Connector/J JAR if not using Maven dependency)
```

Java package declarations must match the package paths exactly. Keep all five JSP files directly inside `src/main/webapp`.

2. MySQL terminal setup

If the database/user have not been created, first log in as a MySQL administrator (for example, root):

```
mysql -u root -p
```

At the MySQL prompt, run the following once to create the database and application account:

```
CREATE DATABASE IF NOT EXISTS student;
CREATE USER IF NOT EXISTS 'student'@'localhost' IDENTIFIED BY 'student';
GRANT ALL PRIVILEGES ON student.* TO 'student'@'localhost';
FLUSH PRIVILEGES;
EXIT;
```

Then log in with the same credentials configured in `DBUtil.java`. When asked for the password, type `student`:

```
mysql -u student -p
```

At the `mysql>` prompt, select the database, create the table and verify it:

```
USE student;
CREATE TABLE IF NOT EXISTS students (
    roll_no INT PRIMARY KEY,
    name VARCHAR(150),
    email VARCHAR(150),
    age INT
);
DESC students;
SELECT * FROM students;
```

Use `mysql -u student -p`; enter the password when prompted. Do not type the password as a SQL statement.

3. Eclipse and driver checklist

• Create/open the Dynamic Web Project using the existing src/main/java and src/main/webapp layout.

• Put the three Java files in the packages shown above.

• Put the five JSP files directly under src/main/webapp.

• Use MySQL Connector/J 8.x or later (driver class: com.mysql.cj.jdbc.Driver). For a non-Maven project, add the JAR under src/main/webapp/WEB-INF/lib and ensure it is included in the deployed app.

• Start MySQL, then run index.jsp on the configured Tomcat server.

Example URL when the context path is StudentApp and Tomcat uses port 8080:

```
http://localhost:8080/StudentApp/index.jsp
```

4. Expected output (examples)

Create: Student added successfully!
Read: displays each record as Roll No, Name, Email and Age.
Update: Student updated successfully!
Delete: Student deleted successfully!

Example data is illustrative; the page displays records stored in your database.

5. Student.java

Location: `src/main/java/com/student/model/Student.java`

```
package com.student.model;

public class Student {
    private int rollNo;
    private String name;
    private String email;
    private int age;

    public Student(int rollNo, String name, String email, int age) {
        this.rollNo = rollNo;
        this.name = name;
        this.email = email;
        this.age = age;
    }

    public int getRollNo() { return rollNo; }
    public String getName() { return name; }
    public String getEmail() { return email; }
    public int getAge() { return age; }

    @Override
    public String toString() {
        return "Student{" +
               "rollNo=" + rollNo +
               ", name='" + name + '\'' +
               ", email='" + email + '\'' +
               ", age=" + age +
               '}';
    }
}
```

6. DBUtil.java

Location: `src/main/java/com/student/util/DBUtil.java`

```
package com.student.util;

import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.SQLException;

public class DBUtil {
    private static final String URL =
        "jdbc:mysql://localhost:3306/student?serverTimezone=UTC";
    private static final String USER = "student";
    private static final String PASSWORD = "student";

    public static Connection getConnection() throws SQLException {
        try {
            Class.forName("com.mysql.cj.jdbc.Driver");
        } catch (ClassNotFoundException e) {
            throw new SQLException("MySQL JDBC driver not found.", e);
        }

        return DriverManager.getConnection(URL, USER, PASSWORD);
    }
}
```

7. StudentDAO.java

Location: `src/main/java/com/student/dao/StudentDAO.java`

```
package com.student.dao;

import com.student.model.Student;
import com.student.util.DBUtil;
import java.sql.*;
import java.util.ArrayList;
import java.util.List;

public class StudentDAO {

    public void addStudent(Student s) throws SQLException {
        String sql = "INSERT INTO students(roll_no, name, email, age) VALUES(?, ?, ?, ?)";

        try (Connection con = DBUtil.getConnection();
             PreparedStatement ps = con.prepareStatement(sql)) {

            ps.setInt(1, s.getRollNo());
            ps.setString(2, s.getName());
            ps.setString(3, s.getEmail());
            ps.setInt(4, s.getAge());
            ps.executeUpdate();
        }
    }

    public List<Student> getAllStudents() throws SQLException {
        List<Student> students = new ArrayList<>();

        try (Connection con = DBUtil.getConnection();
             Statement st = con.createStatement();
             ResultSet rs = st.executeQuery(
                 "SELECT * FROM students ORDER BY roll_no")) {

            while (rs.next()) {
                students.add(new Student(
                    rs.getInt("roll_no"),
                    rs.getString("name"),
                    rs.getString("email"),
                    rs.getInt("age")
                ));
            }
        }

        return students;
    }

    public Student getStudentByRollNo(int rollNo) throws SQLException {
        String sql = "SELECT * FROM students WHERE roll_no = ?";

        try (Connection con = DBUtil.getConnection();
             PreparedStatement ps = con.prepareStatement(sql)) {

            ps.setInt(1, rollNo);

            try (ResultSet rs = ps.executeQuery()) {
                if (rs.next()) {
                    return new Student(
                        rs.getInt("roll_no"),
                        rs.getString("name"),
                        rs.getString("email"),
                        rs.getInt("age")
                    );
                }
            }
        }

        return null;
    }

    public boolean updateStudent(Student s) throws SQLException {
        String sql =
            "UPDATE students SET name=?, email=?, age=? WHERE roll_no=?";

        try (Connection con = DBUtil.getConnection();
             PreparedStatement ps = con.prepareStatement(sql)) {

            ps.setString(1, s.getName());
            ps.setString(2, s.getEmail());
            ps.setInt(3, s.getAge());
            ps.setInt(4, s.getRollNo());

            return ps.executeUpdate() > 0;
        }
    }

    public boolean deleteStudent(int rollNo) throws SQLException {
        String sql = "DELETE FROM students WHERE roll_no = ?";

        try (Connection con = DBUtil.getConnection();
             PreparedStatement ps = con.prepareStatement(sql)) {

            ps.setInt(1, rollNo);
            return ps.executeUpdate() > 0;
        }
    }
}
```

8. index.jsp

Location: `src/main/webapp/index.jsp`

```
<%@ page contentType="text/html;charset=UTF-8" %>
<html>
<body>
    <h2>Student Management</h2>

    <a href="add-student.jsp">Add Student</a><br><br>
    <a href="list-student.jsp">View Students</a><br><br>
    <a href="delete-student.jsp">Delete Student by Roll No</a>
</body>
</html>
```

9. add-student.jsp

Location: `src/main/webapp/add-student.jsp`

```
<%@ page contentType="text/html;charset=UTF-8" %>
<%@ page import="com.student.dao.StudentDAO,com.student.model.Student" %>
<html>
<body>
    <h2>Add Student</h2>

    <form action="add-student.jsp" method="post">
        Roll No: <input type="number" name="rollNo" required><br><br>
        Name: <input type="text" name="name" required><br><br>
        Email: <input type="email" name="email" required><br><br>
        Age: <input type="number" name="age" required><br><br>
        <input type="submit" value="Add Student">
    </form>

    <%
        if ("POST".equalsIgnoreCase(request.getMethod())) {
            try {
                int rollNo = Integer.parseInt(request.getParameter("rollNo"));
                String name = request.getParameter("name");
                String email = request.getParameter("email");
                int age = Integer.parseInt(request.getParameter("age"));

                Student s = new Student(rollNo, name, email, age);
                new StudentDAO().addStudent(s);

                out.println("<p>Student added successfully!</p>");
            } catch (Exception e) {
                out.println("<p>Could not add student: " + e.getMessage() + "</p>");
            }
        }
    %>

    <br><a href="list-student.jsp">View Students</a><br><br>
    <a href="index.jsp">Back to Home</a>
</body>
</html>
```

10. list-student.jsp

Location: `src/main/webapp/list-student.jsp`

```
<%@ page contentType="text/html;charset=UTF-8" %>
<%@ page import="com.student.dao.StudentDAO,com.student.model.Student,java.util.List" %>
<html>
<body>
    <h2>Student List</h2>

    <%
        try {
            List<Student> students = new StudentDAO().getAllStudents();

            if (students.isEmpty()) {
                out.println("<p>No students found.</p>");
            }

            for (Student s : students) {
    %>
                Roll No: <%= s.getRollNo() %><br>
                Name: <%= s.getName() %><br>
                Email: <%= s.getEmail() %><br>
                Age: <%= s.getAge() %><br>

                <a href="update-student.jsp?rollNo=<%= s.getRollNo() %>">Update</a> |
                <a href="delete-student.jsp?rollNo=<%= s.getRollNo() %>">Delete</a>

                <hr>
    <%
            }
        } catch (Exception e) {
            out.println("<p>Could not load students: " + e.getMessage() + "</p>");
        }
    %>

    <a href="add-student.jsp">Add Student</a><br><br>
    <a href="index.jsp">Back to Home</a>
</body>
</html>
```

11. update-student.jsp

Location: `src/main/webapp/update-student.jsp`

```
<%@ page contentType="text/html;charset=UTF-8" %>
<%@ page import="com.student.dao.StudentDAO,com.student.model.Student" %>
<html>
<body>
    <h2>Update Student</h2>

    <%
        StudentDAO dao = new StudentDAO();
        Student student = null;
        String message = "";
        boolean post = "POST".equalsIgnoreCase(request.getMethod());

        try {
            if (post) {
                int rollNo = Integer.parseInt(request.getParameter("rollNo"));
                String name = request.getParameter("name");
                String email = request.getParameter("email");
                int age = Integer.parseInt(request.getParameter("age"));

                Student s = new Student(rollNo, name, email, age);

                if (dao.updateStudent(s))
                    message = "Student updated successfully!";
                else
                    message = "No student found with that roll number.";
            } else {
                String id = request.getParameter("rollNo");

                if (id != null && !id.trim().isEmpty())
                    student = dao.getStudentByRollNo(Integer.parseInt(id));
            }
        } catch (Exception e) {
            message = "Could not update student: " + e.getMessage();
        }

        if (!message.isEmpty()) {
            out.println("<p>" + message + "</p>");
        }

        if (!post) {
            if (student != null) {
    %>
                <form action="update-student.jsp" method="post">
                    Roll No: <%= student.getRollNo() %><br><br>

                    <input type="hidden" name="rollNo"
                           value="<%= student.getRollNo() %>">

                    Name: <input type="text" name="name"
                        value="<%= student.getName() %>" required><br><br>

                    Email: <input type="email" name="email"
                        value="<%= student.getEmail() %>" required><br><br>

                    Age: <input type="number" name="age"
                        value="<%= student.getAge() %>" required><br><br>

                    <input type="submit" value="Update Student">
                </form>
    <%
            } else {
                out.println("<p>Student not found. Open the student list and select Update.</p>");
            }
        }
    %>

    <br><a href="list-student.jsp">Back to Student List</a><br><br>
    <a href="index.jsp">Back to Home</a>
</body>
</html>
```

12. delete-student.jsp

Location: `src/main/webapp/delete-student.jsp`

```
<%@ page contentType="text/html;charset=UTF-8" %>
<%@ page import="com.student.dao.StudentDAO" %>
<html>
<body>
    <h2>Delete Student</h2>

    <%
        String rollNoValue = request.getParameter("rollNo");
        String message = "";

        if ("POST".equalsIgnoreCase(request.getMethod())) {
            try {
                int rollNo = Integer.parseInt(rollNoValue);

                if (new StudentDAO().deleteStudent(rollNo))
                    message = "Student deleted successfully!";
                else
                    message = "No student found with that roll number.";
            } catch (Exception e) {
                message = "Could not delete student: " + e.getMessage();
            }
        }

        if (!message.isEmpty()) {
            out.println("<p>" + message + "</p>");
        }

        if (rollNoValue == null)
            rollNoValue = "";
    %>

    <form action="delete-student.jsp" method="post">
        Enter Roll No:
        <input type="number" name="rollNo"
               value="<%= rollNoValue %>" required><br><br>
        <input type="submit" value="Delete Student">
    </form>

    <br><a href="list-student.jsp">Back to Student List</a><br><br>
    <a href="index.jsp">Back to Home</a>
</body>
</html>
```

13. database.sql

```
CREATE DATABASE IF NOT EXISTS student;
CREATE USER IF NOT EXISTS 'student'@'localhost' IDENTIFIED BY 'student';
GRANT ALL PRIVILEGES ON student.* TO 'student'@'localhost';
FLUSH PRIVILEGES;

USE student;
CREATE TABLE IF NOT EXISTS students (
    roll_no INT PRIMARY KEY,
    name VARCHAR(150),
    email VARCHAR(150),
    age INT
);
```

14. Quick verification before the lab

• Run DESC students; and SELECT * FROM students; in MySQL to confirm the table exists.

• Add one sample student through the web form and verify it with SELECT * FROM students;.

• Open View Students; check that Update pre-fills the selected student.

• Update the student and verify the changed values in MySQL.

• Delete that test record and verify it no longer appears.

• If JDBC driver not found, check the Connector/J deployment location. If access denied, check DBUtil credentials and MySQL grants. If the page is 404, check project context path and JSP location.

Code review status: the three Java source files compiled successfully. JSP syntax, servlet container configuration, database access and form flows remain to be verified in your live Eclipse/Tomcat/MySQL setup.